Department of CSE(AIML), PES University
UE26CS151A -Python for Computational Problem Solving
Laboratory Week 6

Note: Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. The observation record must be signed by the faculty in charge. For every laboratory session, the observation component carries **marks**.

Problem Statements

Q1. Display Lines Starting with a Given Letter
Write a program to read a text file and display all lines that start with a letter entered by the user.
Q2. Find the Student with the Highest Marks Using a Dictionary
Write a program to store student names and marks in a dictionary and display the name and marks of the student who scored the highest marks.
Q3. Calculate the Total Amount for Each Order from a CSV File
Write a program to read product names, quantities, and prices from week_6_set3_q3.csv and display the total amount for each order by multiplying its quantity by its price.
Q4. Count the Frequency of Each Word in a Text File
Write a program to read a text file and use a dictionary to display each unique word and its frequency.
Q5. Find Products Within a Given Price Range from a CSV File
Write a program to read product names and prices from week_6_set3_q5.csv and display the products whose prices fall within a user-entered minimum and maximum price.
Q6. Merge Two Dictionaries Containing Student Marks
Write a program to create two dictionaries containing student names and marks, merge them into a single dictionary, and display the result. If a student appears in both dictionaries, retain the marks from the second dictionary.
Q7. Find the Longest Word in a Text File
Write a program to read a text file and display the longest word present in it. If multiple words have the same maximum length, display the first such word.
Q8. Store City Names and Populations in a CSV File
Write a Python program that asks the user to enter the names of three cities and their populations. Store the data in a list of lists and write the results into a CSV file named week_6_set3_q8.csv.
Solutions
1. Display Lines Starting with a Given Letter

```
file = open("cities.txt", "w")
file.write("Bengaluru is a large city\n")
file.write("Delhi is the capital of India\n")
file.write("Mumbai is known for business\n")
file.write("Bengaluru has many colleges\n")
file.close()
```

```
file_name = input("Enter file name: ")
letter = input("Enter starting letter: ")

file = open(file_name, "r")

print("Lines starting with", letter, ":")

for line in file:
    if line.lower().startswith(letter.lower()):
print(line, end="")

file.close()
```

Output 1:

```
Enter file name: cities.txt
Enter starting letter: B
Lines starting with B :
Bengaluru is a large city
Bengaluru has many colleges
```

Output 2:

```
Enter file name: cities.txt
Enter starting letter: M
Lines starting with M :
Mumbai is known for business
```

2. Find the Student with the Highest Marks Using a Dictionary

```
students = {}

n = int(input("Enter number of students: "))

for i in range(n):
    name = input("Enter student name: ")
    marks = int(input("Enter marks: "))
    students[name] = marks

highest_name = ""
highest_marks = -1

for name in students:
    if students[name] >highest_marks:
highest_marks = students[name]
highest_name = name

print("Student with highest marks:", highest_name)
```

```
print("Highest marks:", highest_marks)
```

Output 1:

```
Enter number of students: 3
Enter student name: Aarav
Enter marks: 78
Enter student name: Diya
Enter marks: 92
Enter student name: Kabir
Enter marks: 85
Student with highest marks: Diya
Highest marks: 92
```

Output 2:

```
Enter number of students: 3
Enter student name: Meera
Enter marks: 76
Enter student name: Rohan
Enter marks: 88
Enter student name: Ananya
Enter marks: 95
Student with highest marks: Ananya
Highest marks: 95
```

3. Calculate the Total Amount for Each Order from a CSV File

```
import csv

file = open("week_6_set3_q3.csv", "r")
reader = csv.DictReader(file)

print("Total amount for each order:")

for row in reader:
    name = row["Product Name"]
    quantity = int(row["Quantity"])
    price = float(row["Price"])

    total = quantity * price

print(name, ":", total)

file.close()
```

Output:

```
Total amount for each order:
Notebook : 135.0
```

Pen : 120.0
Backpack : 1300.0
Marker : 140.0
Eraser : 30.0

4. Count the Frequency of Each Word in a Text File

```
file = open("words.txt", "w")
file.write("python is easy python is useful")
file.close()

file_name = input("Enter file name: ")

file = open(file_name, "r")
text = file.read()
file.close()

words = text.split()
frequency = {}

for word in words:
    if word in frequency:
        frequency[word] += 1
    else:
        frequency[word] = 1

print("Word frequencies:")

for word in frequency:
print(word, ":", frequency[word])
```

Output 1:

Enter file name: words.txt
Word frequencies:
python : 2
is : 2
easy : 1
useful : 1

Output 2:

Enter file name: words.txt
Word frequencies:
python : 2
is : 2
easy : 1
useful : 1

5. Find Products Within a Given Price Range from a CSV File

```
import csv

file = open("week_6_set3_q5.csv", "r")
reader = csv.DictReader(file)

minimum = float(input("Enter minimum price: "))
maximum = float(input("Enter maximum price: "))

print("Products within the price range:")

for row in reader:
    name = row["Product Name"]
    price = float(row["Price"])

    if minimum <= price <= maximum:
print(name, ":", price)

file.close()
```

Output 1:

```
Enter minimum price: 50
Enter maximum price: 200
Products within the price range:
Water Bottle : 180.0
Pencil Box : 95.0
```

Output 2:

```
Enter minimum price: 10
Enter maximum price: 50
Products within the price range:
Notebook : 45.0
Pen : 12.0
Marker : 35.0
```

6. Merge Two Dictionaries Containing Student Marks

```
students1 = {}
students2 = {}

n1 = int(input("Enter number of students in first dictionary: "))

for i in range(n1):
    name = input("Enter student name: ")
    marks = int(input("Enter marks: "))
    students1[name] = marks
```

```
n2 = int(input("Enter number of students in second dictionary: "))

for i in range(n2):
    name = input("Enter student name: ")
    marks = int(input("Enter marks: "))
    students2[name] = marks

students1.update(students2)

print("Merged dictionary:", students1)
```

Output 1:

```
Enter number of students in first dictionary: 2
Enter student name: Aarav
Enter marks: 78
Enter student name: Diya
Enter marks: 85
Enter number of students in second dictionary: 2
Enter student name: Diya
Enter marks: 92
Enter student name: Kabir
Enter marks: 88
Merged dictionary: {'Aarav': 78, 'Diya': 92, 'Kabir': 88}
```

Output 2:

```
Enter number of students in first dictionary: 2
Enter student name: Meera
Enter marks: 76
Enter student name: Rohan
Enter marks: 80
Enter number of students in second dictionary: 1
Enter student name: Ananya
Enter marks: 95
Merged dictionary: {'Meera': 76, 'Rohan': 80, 'Ananya': 95}
```

7. Find the Longest Word in a Text File

```
file = open("story.txt", "w")
file.write("Python programming develops logical thinking skills")
file.close()

file_name = input("Enter file name: ")

file = open(file_name, "r")
words = file.read().split()
```

```
file.close()

longest = ""

for word in words:
    if len(word) >len(longest):
        longest = word

if len(words) > 0:
print("Longest word:", longest)
print("Length:", len(longest))
else:
print("File contains no words")
```

Output 1:

```
Enter file name: story.txt
Longest word: programming
Length: 11
```

Output 2:

```
Enter file name: story.txt
Longest word: programming
Length: 11
```

8. Store City Names and Populations in a CSV File

```
import csv

cities = []

for i in range(3):
    name = input("Enter city name: ")
    population = int(input("Enter population: "))
cities.append([name, population])

file = open("week_6_set3_q8.csv", "w", newline="")
writer = csv.writer(file)

writer.writerow(["City", "Population"])
writer.writerows(cities)

file.close()

print("Data written to week_6_set3_q8.csv")

file = open("week_6_set3_q8.csv", "r")
```

```
reader = csv.reader(file)

for row in reader:
    print(row[0], row[1])

file.close()
```

Output:

```
Enter city name: Bengaluru
Enter population: 12000000
Enter city name: Delhi
Enter population: 18000000
Enter city name: Mumbai
Enter population: 20000000
Data written to week_6_set3_q8.csv
City Population
Bengaluru 12000000
Delhi 18000000
Mumbai 20000000
```

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